Polynomial Equations & Factoring: Area Model & [EQN] Solver

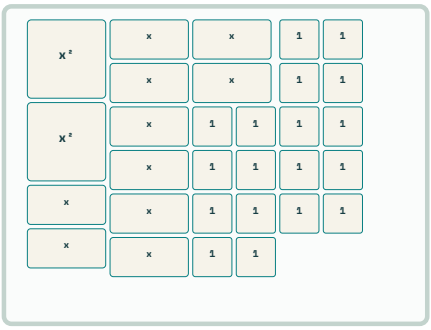
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.HSA.APR.A.1

Student Name: _____ Date: _____ **Model: SC-32** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-32 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Algebra Tiles

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Arrange 1 x^2 tile (40x40mm), 5 x-tiles (40x10mm), and 6 unit squares (10x10mm) into solid rectangle.

Sketch Model on Paper:
Draw 2x2 area matrix with dimensions (x + 2) and (x + 3).

3. Step-by-Step LCD Stack Trace (SC-32)

MODE:	EQN	Polynomial Solver	
1.	1	X: a = 1	Key coeff a (x^2) int
2.	ENTER	Y: 1 | X: 1	Stack lift
3.	5	Y: 1 | X: b = 5	Key coeff b (x) into
4.	ENTER	Z: 1 | Y: 5 | X: 5	Stack lift

EXPECTED LCD X: Roots: x1 = -2.0000, x2 =

1 **Find roots of $x^2 + 6x + 8 = 0$ using [EQN].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

2 **Find roots of $x^2 - 6x + 8 = 0$ using [EQN].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

3 **Find roots of $x^2 - 9 = 0$ using [EQN].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

4 **Find roots of $x^2 + 6x + 9 = 0$ (perfect square).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

Station Diagnostic & Solution Key: Chapter 7

Polynomial Equations & Factoring: Area Model & [EQN] Solver • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Tactile Algebra Stage & Polynomial Tiles

CAD Blueprint: algebra_tiles

Target Device: SC-32 Scientific

CAD / STL Source File:

Education/scad/algebra_factoring_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal x^2 Tiles, Amber x-Rods & White Units (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Find roots of $x^2 + 6x + 8 = 0$ using [EQN].** Standard: CCSS.MATH.HSA.APR.A.1	**x1 = -2, x2 = -4**	[1] [ENTER] [6] [ENTER] [8] [EQN]
2	**Find roots of $x^2 - 6x + 8 = 0$ using [EQN].** Standard: CCSS.MATH.HSA.APR.A.1	**x1 = 2, x2 = 4**	[1] [ENTER] [6] [+/-] [ENTER] [8] [EQN
3	**Find roots of $x^2 - 9 = 0$ using [EQN].** Standard: CCSS.MATH.HSA.APR.A.1	**x1 = 3, x2 = -3**	[1] [ENTER] [0] [ENTER] [9] [+/-] [EQN
4	**Find roots of $x^2 + 6x + 9 = 0$ (perfect squar** Standard: CCSS.MATH.HSA.APR.A.1	**x1 = -3 (Repeated)**	[1] [ENTER] [6] [ENTER] [9] [EQN]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Roots on LCD have opposite signs from binomial factors: root r = -2 corresponds to factor (x + 2).

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-32, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.